SKILLCLAW FRAMEWORK

SkillClaw: Let Skills Evolve Collectively with Agentic Evolver

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

"Building a continuously improving skill ecosystem without manual curation"

April 20, 2026

Source: arxiv.org/abs/2604.08377

PROBLEM STATEMENT

Agent Skills Remain Static After Deployment — Users Rediscover Solutions Independently

Static Skill Libraries

No Post-Deployment Growth

Current LLM agents rely on reusable skills installed from centralized hubs.

These skills remain completely static and do not evolve through usage and interaction.

Bottleneck

Isolated Recoveries

Solutions Rarely Persist

Solutions discovered during agent interaction rarely persist beyond individual sessions.

Similar tasks recur across users over time, repeating the same patterns of failure and recovery.

Inefficiency

Wasted Effort

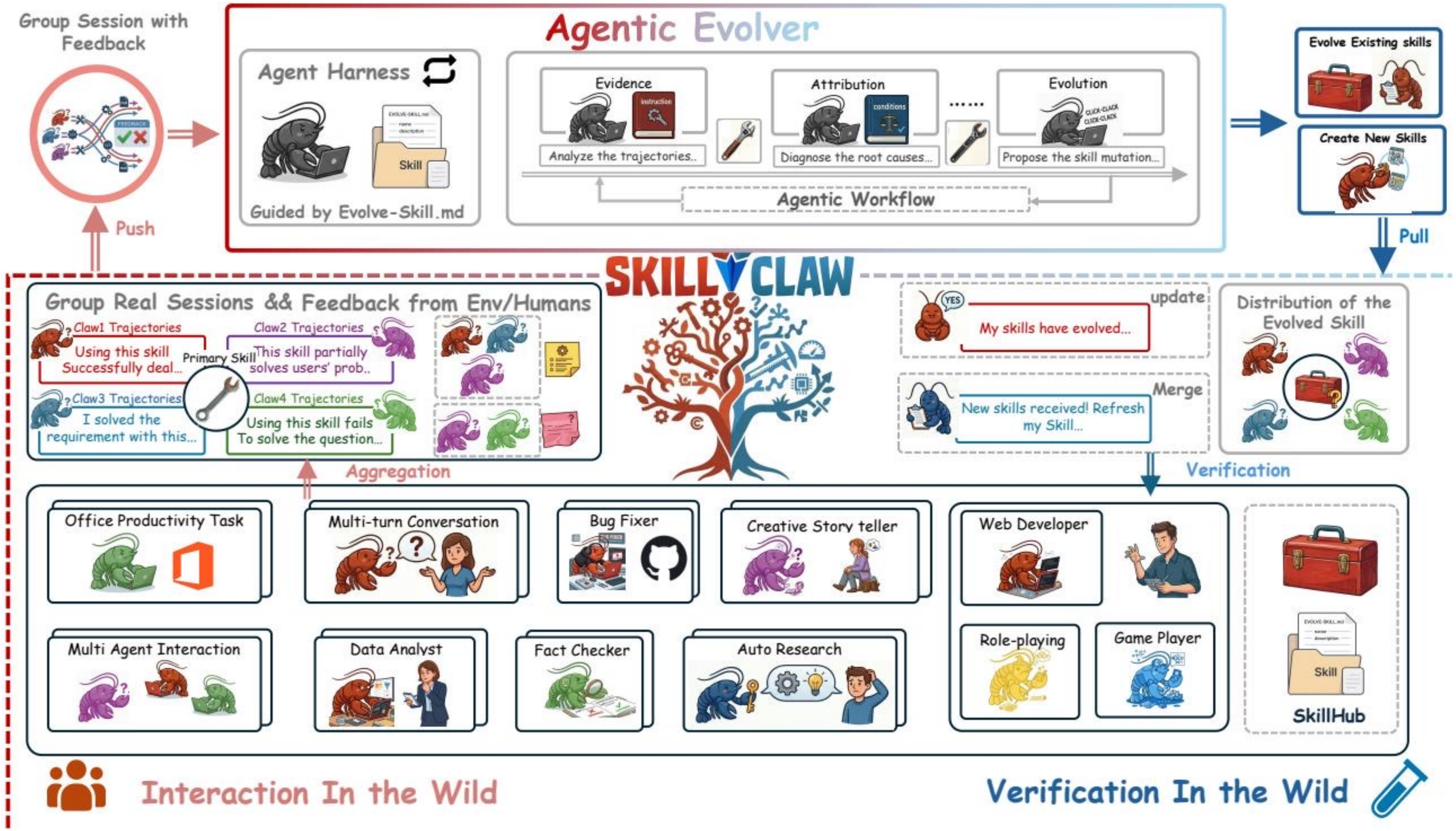
No System-Level Knowledge

Each user is forced to rediscover solutions independently.

This prevents knowledge from accumulating at the system level, keeping the agent ecosystem in a primitive state.

Consequence

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution



Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

Stage 1: Evidence

Analyze Behavioral Patterns

The evolver systematically analyzes aggregated interaction trajectories.

It identifies consistent patterns of failures, recovery paths, and successes across multiple users.

Analysis

Stage 2: Attribution

Diagnose Root Causes

Agentic reasoning is applied to diagnose the exact reasons behind observed behaviors.

Pinpoints specific skill flaws or missing steps responsible for the execution errors.

Diagnosis

Stage 3: Evolution

Propose Skill Mutations

Driven by open-ended reasoning, not predefined update rules.

Generates concrete skill mutations: either refining an existing skill's logic or creating entirely new missing skills.

Mutation

AGGREGATION

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Diverse Context Signals

Complementary Behavioral Data

Trajectories from different users across compatible Claw-style systems:

OpenClaw | CoPaw | IronClaw | PicoClaw | ZeroClaw | NanoClaw | NemoClaw

They exercise the same skill under diverse contexts, producing complementary signals of success patterns and failure modes.

The Single User Limit

Idiosyncratic vs Generalizable

A single user rarely generates enough signal to improve the system.

Without aggregation, it is impossible to separate a generalizable skill improvement from a highly idiosyncratic local fix.

Aggregation reveals true behavioral boundaries

Case Study: Multimodal Creative Pipeline Skill — 6 Days of Evolution Attempts

Day	Candidate Skill	Skill Function	Validator	Action
1	validate-tmp-workspace-inputs	Check inputs & environment setup	Accept	Upgrade to new best pool
2	multimodal-input-validation	Input validation & env init	Reject	Keep current best pool
3	multimodal-creative-pipeline	Pipeline (extract from PDF/media)	Reject	Keep current best pool
4	multimodal-creative (impr.)	Added image class, structured val	Reject	Keep current best pool
5	creative (impr.) + validate-req	Pipeline + per-file fail-fast	Reject	Keep current best pool
6	multimodal-creative (cand.)	Extended PDF-to-poster paths	Reject	Not admitted to next cycle

Only 1 of 6 candidates was accepted illustrating the conservative verification process that prevents regressions.

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

Day	Candidate Skill	Change Summary	Validator	Action
1	git-push-with-auth-fallback	Safe fallback on push failure	Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Unified filenames, reduced drift	Accept	Keep updated best pool
3	fallback + clone-to-directory	Auth paths & secrets audit	Accept	Keep current best pool
4	(none)	Retest; confirmed current is best	Accept	Continue same best pool
5	git-push-with-auth-fallback	Push hang treated as auth fail	Reject	Keep current best pool
6	git-push-with-auth-fallback	Added identity config	Reject	Not admitted to next cycle

3 of 6 days yielded accepted updates showing successful iterative refinement before reaching a plateau.

CONCLUSION

Key Takeaways: Three Properties That Distinguish SkillClaw

1

Collective Evolution

Shared Ecosystem Growth

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem.

Collective

2

Fully Automatic

No Manual Curation

Skill evolution is driven entirely by runtime interaction without manual curation or explicit user intervention.

Automatic

3

Agentic Paradigm

Open-Ended Reasoning

Skill updates are produced through open-ended reasoning over evidence rather than predefined update rules.

Agentic